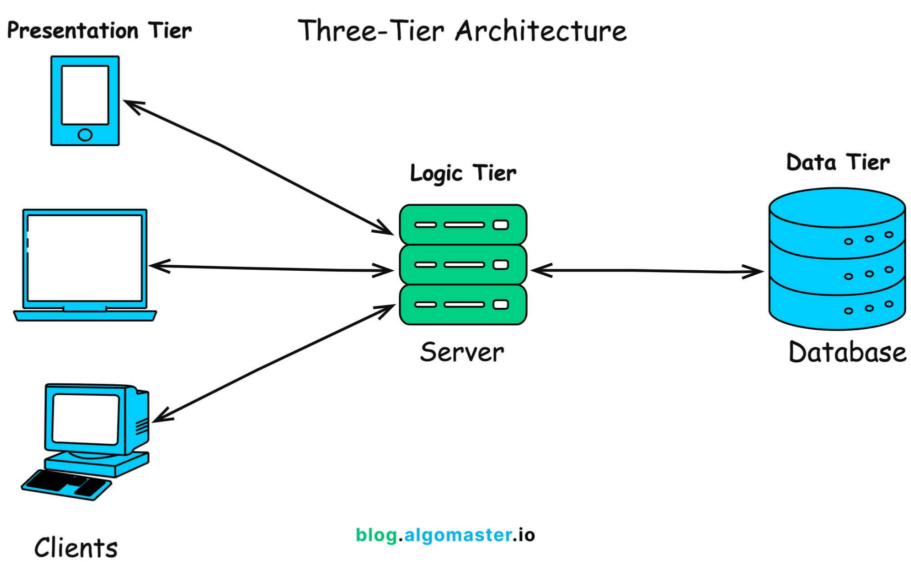
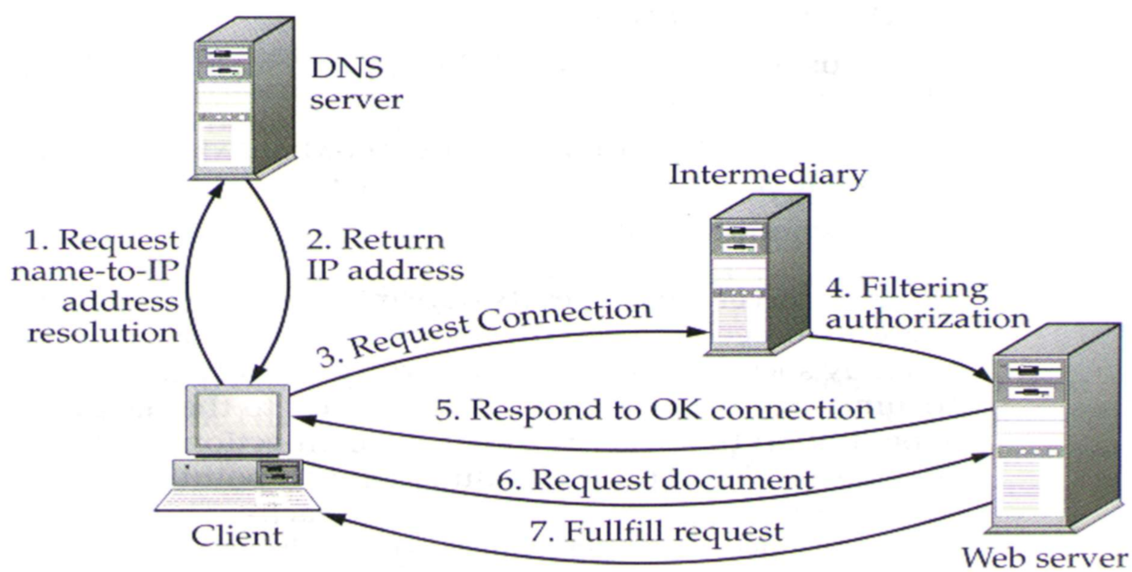
Assignment 1

Assignment Name: Internet Explorer

1. Client-Server Architecture





Definition

Client-server architecture is a network model in which multiple clients (users/devices) request services and resources from centralized servers.

Components

1. **Client**
   - A device such as a computer, smartphone, or tablet.
   - Uses a web browser to send requests.
   - Example: Chrome, Edge, Firefox.
2. **Server**
   - A powerful computer that stores websites and data.
   - Handles requests and sends responses.
   - Types include web servers, database servers, and application servers.
3. **Internet (Network)**
   - The communication medium between client and server.
   - Uses protocols like HTTP/HTTPS.

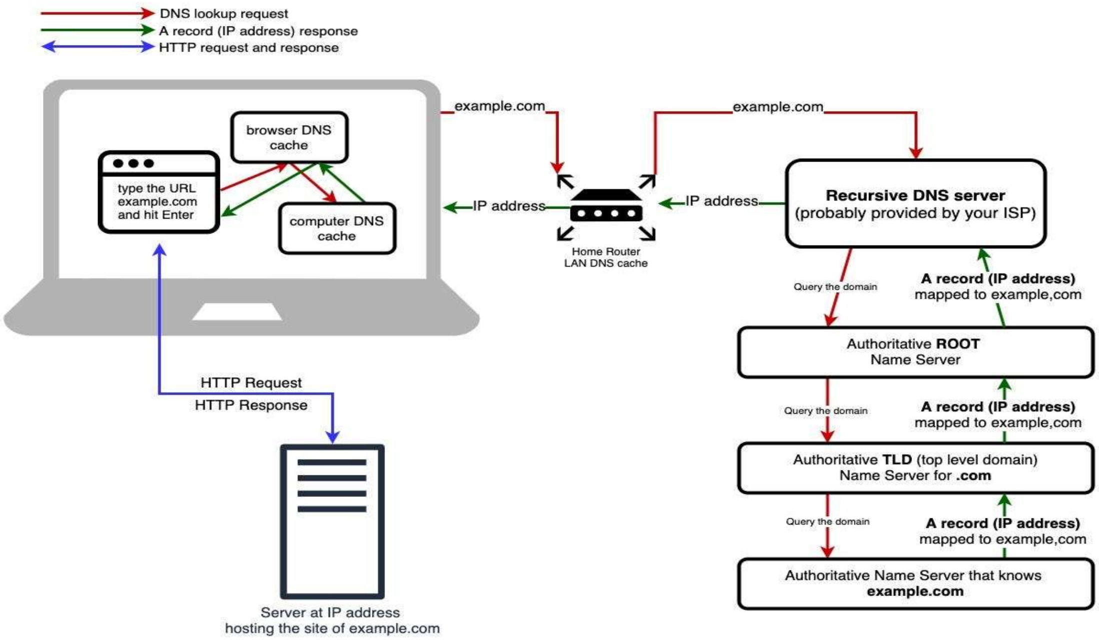
Working

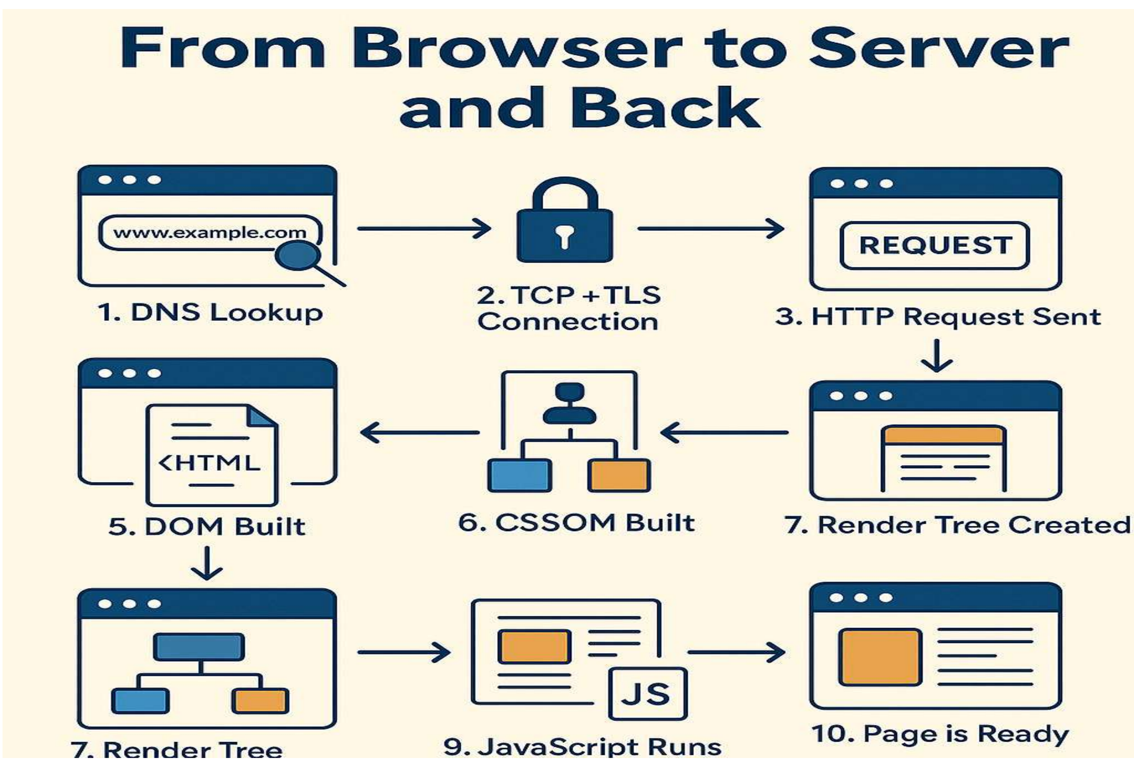
- The client sends a request to the server.
- The server receives and processes the request.
- The server sends back a response.
- The client displays the result.

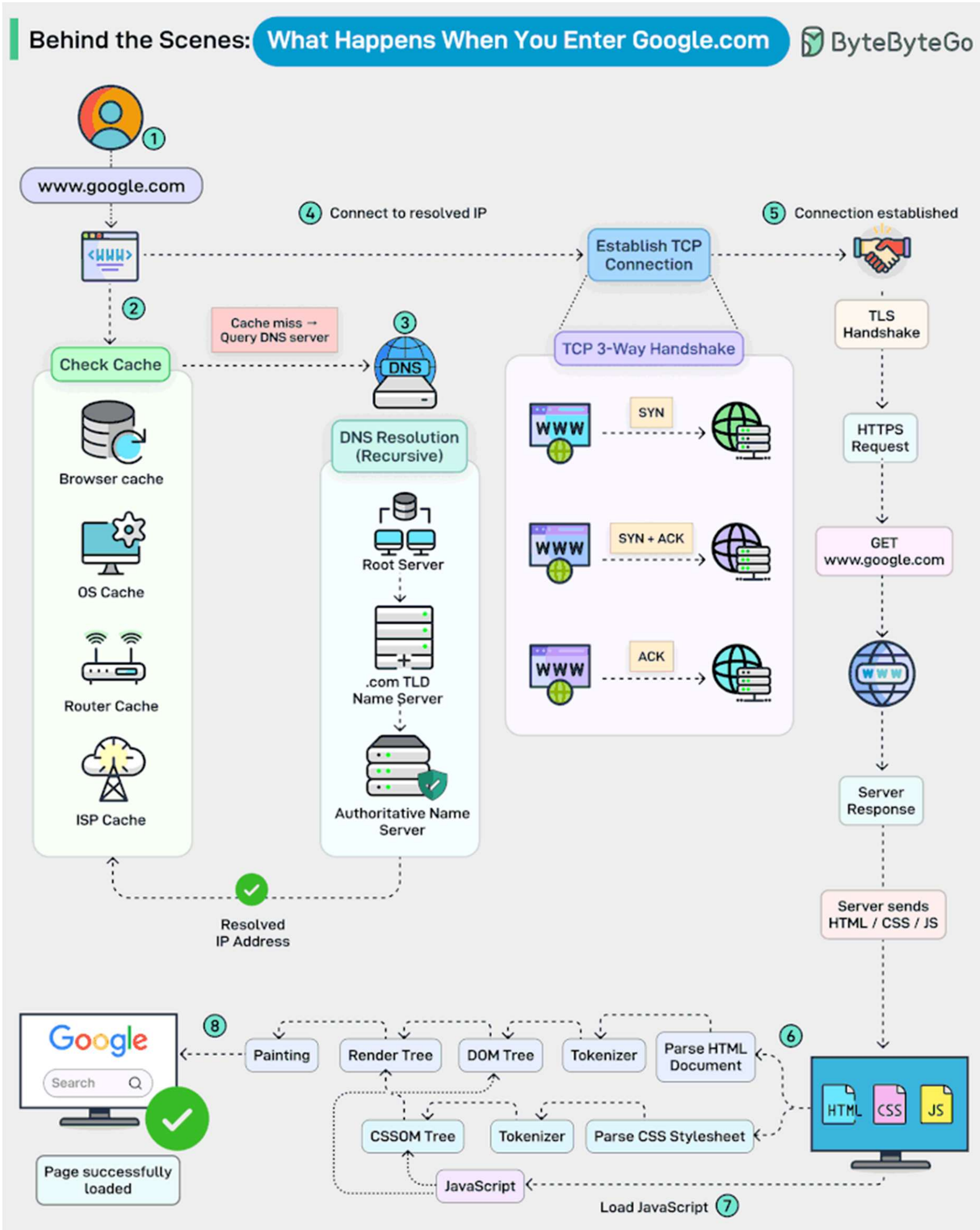
Key Features

- **Centralized Control:** Data is stored on servers.
- **Scalability:** Multiple clients can connect simultaneously.
- **Security:** Servers can control access and permissions.

2. What Happens When You Open a Website


DNS lookup request
A record (IP address) response
HTTP request and response
browser DNS cache
type the URL example.com and hit Enter
computer DNS cache
example.com
IP address
Home Router LAN DNS cache
example.com
IP address
Recursive DNS server (probably provided by your ISP)
Query the domain
A record (IP address) mapped to example,com
Authoritative ROOT Name Server
Query the domain
A record (IP address) mapped to example,com
Authoritative TLD (top level domain) Name Server for .com
Query the domain
A record (IP address) mapped to example,com
Authoritative Name Server that knows example.com
HTTP Request
HTTP Response
Server at IP address hosting the site of example.com


From Browser to Server and Back
www.example.com
1. DNS Lookup
2. TCP +TLS Connection
REQUEST
3. HTTP Request Sent
<HTML
5. DOM Built
6. CSSOM Built
7. Render Tree Created
7. Render Tree
JS
9. JavaScript Runs
10. Page is Ready



Step-by-Step Explanation

1. **Entering the URL**
    - The user types a website address (URL) into the browser.
2. **DNS Lookup**
    - The browser contacts a DNS server.

    - The domain name is translated into an IP address.
    - Example: google.com → 142.x.x.x
3. **TCP Connection**
    - The browser establishes a connection with the server using TCP (Transmission Control Protocol).
4. **TLS Handshake (for HTTPS)**
    - If the website uses HTTPS, a secure connection is established.
    - Data becomes encrypted.
5. **HTTP Request**
    - The browser sends an HTTP request (GET/POST) to the server.
6. **Server Processing**
    - The server processes the request.
    - It may interact with a database to fetch data.
7. **HTTP Response**
    - The server sends back:
        - HTML (structure)
        - CSS (design)
        - JavaScript (functionality)
8. **Rendering the Webpage**
    - The browser interprets the files.
    - It builds the page and displays it.

Important Terms

- **URL (Uniform Resource Locator):** Address of a website.
- **DNS (Domain Name System):** Converts domain names into IP addresses.
- **HTTP/HTTPS:** Protocols used for communication.
- **IP Address:** Unique identifier of a server.
- **Browser Rendering:** Process of displaying content visually.

3. Flow Summary

```
User → Browser → DNS → TCP/TLS → Server → Response → Browser → Display
```

4. Conclusion

Client-server architecture is the foundation of web communication. Every time a website is accessed, a structured process involving DNS resolution, connection establishment, request handling, and response delivery occurs, allowing the browser to display the webpage accurately.